

Project Methodology

A complete end-to-end retail analytics workflow — synthetic data generation to Power BI dashboards.

Total Sales	Total Profit	Records	Columns
\$12.74M	\$3.19M	1,000+	25

Methodology Phases

01

Data Generation

- Used Faker, NumPy, and Pandas to build 1000+ synthetic retail orders from scratch
- 25 columns covering orders, customers, products, pricing, fulfillment, and inventory
- Embedded auto-reorder business logic: stock below threshold triggers reorder quantity suggestion
- Realistic distributions designed across 4 regions, 3 customer segments, 4 product categories

Tool: Faker, NumPy, Pandas, Random

02

Data Preparation

- Confirmed zero missing values and zero duplicate rows across all 25 columns
- Parsed Order Date and Ship Date from string to datetime (format: %d-%m-%Y)
- Engineered Order Month (Period) and Order Year columns for time series analysis
- Computed Profit Margin % per transaction for comparative analysis

Tool: Pandas (pd.to_datetime, dt.to_period)

Exploratory Data Analysis

- Univariate: value_counts and distribution plots for 6 categorical variables; histogram + boxplot for 8 numerical variables
- Bivariate: groupby aggregations for Sales, Profit, Order Count by Category, Region, and Customer Segment
- Time Series: monthly aggregation of sales, profit, and order count; trend lines with markers
- Correlation: Pearson correlation heatmap across Quantity, Unit Price, Discount, Sales, Cost Price, Profit
- Key finding: discount has weak correlation with profit; cost price and sales are strongly correlated

Tool: Pandas, Matplotlib, Seaborn

Power BI Dashboards

- Dashboard 1 (Performance): KPIs, top products by sales, sales/profit by category, regional pie charts, monthly trend
- Dashboard 2 (Inventory): Stock health, auto-reorder pie (19.4% flagged), reorder quantity trend, supplier table
- Dashboard 3 (Customer): 989 unique customers, profit by segment & region, sales by delivery status, payment mode breakdown
- Interactive slicers added for Region, Category, Payment Mode, and Customer Segment across all dashboards

Tool: Microsoft Power BI

Bivariate Analysis Detail

Dimension	Metric	Top Performer	Insight
Category	Sales Amount	Electronics (\$3.3M)	Highest raw volume
Category	Profit Margin	Grocery (25.8%)	Best margin efficiency
Region	Total Profit	North (\$0.87M)	Highest absolute profit
Region	Sales Share	West (27.2%)	Largest share by sales
Customer Segment	Order Count	Home Office (354)	Largest by volume
Customer Segment	Profit/Order	Corporate	Most efficient per order

Key Insights

Best margin

Grocery

y

Leads in profit margin despite Electronics having higher sales volume

Top region	North	Highest absolute profit at \$0.87M — most efficient region
Low stock	19.4%	Auto-reorder logic flags 194 items proactively before stockout
Top payment	UPI	Most common payment method — digital-first customer behaviour

Python 3.x	Pandas	NumPy	Matplotlib	Seaborn	Faker	Power BI
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